Data Description

All tables contain the `climactor_id` column (the key on which the DataFrames can be merged).

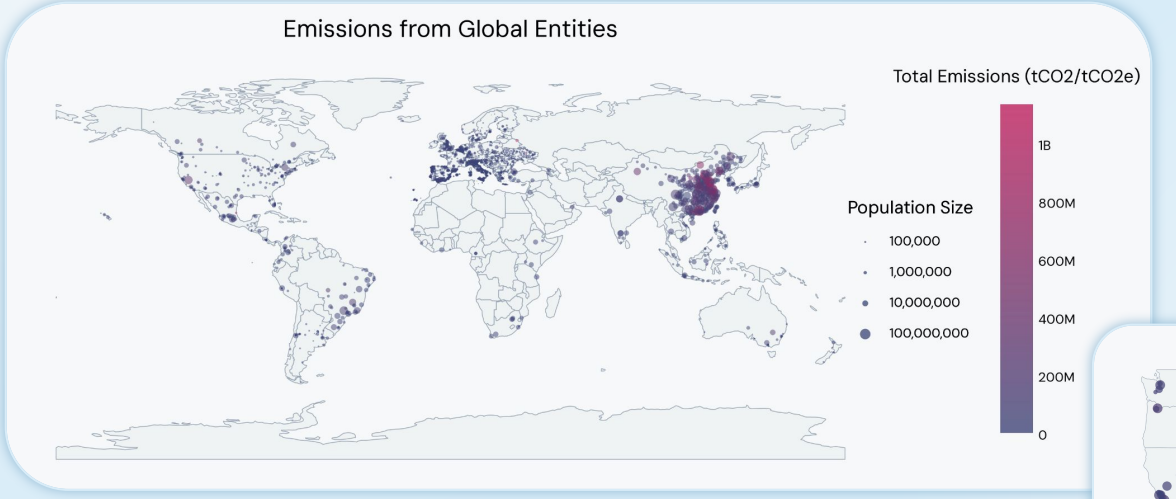
	File	Contents	Important Features*
1)	**`keydict`** (parquet)	Descriptors and geographic locations for each entity, alongside other geographic keys for merging external datasets	`long_lat, area, population`
2)	**`actor`**	Entity identities	`name, iso, entity_type`
3)	**`emissions`**	Measures of emissions across entities and years	`total_emissions, inv_year, inv_unit`
4)	**`emissions by sector`**	Breaks down **`emissions`** into the sectors responsible	`emissions_value, inv_year, inv_unit, sector`
5)	**`targets`**	Entities' emission target types and target years	`target_type, target_year`
6)	**`plans`**	Entities' climate action plans (both as simple categories and detailed descriptions)	`plan_type, plan_adopted_year`
7)	**`initiatives`**	Entities' official climate initiatives	`initiative_name`
8)	**`IPCC plans & enhanced categories`**	Primarily, updated columns for **`keydict`** and **`plans`**, but with fewer rows	

**Determined based on: having relatively few missing values, having relatively few distinct values (if categorical), and generally lending themselves to interpretable visualizations*

Emissions Visualization

After filtering for only the most recent year of emissions data for each entity, I plotted `total_emissions` *on a world map, with point size corresponding to* `population`.

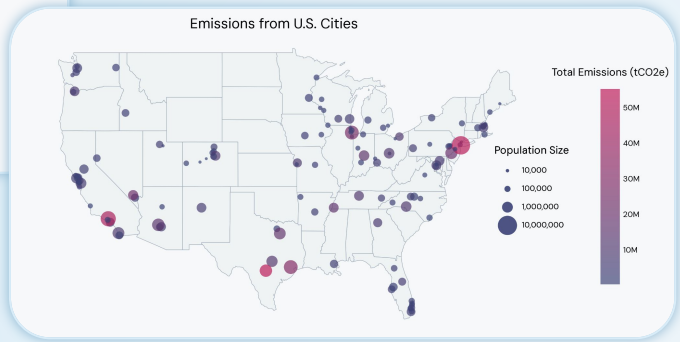
global_emissions.png



`inv_unit` *(unit of emissions) and* `entity_type` *were not filtered, which introduces potential redundancies and biases.*

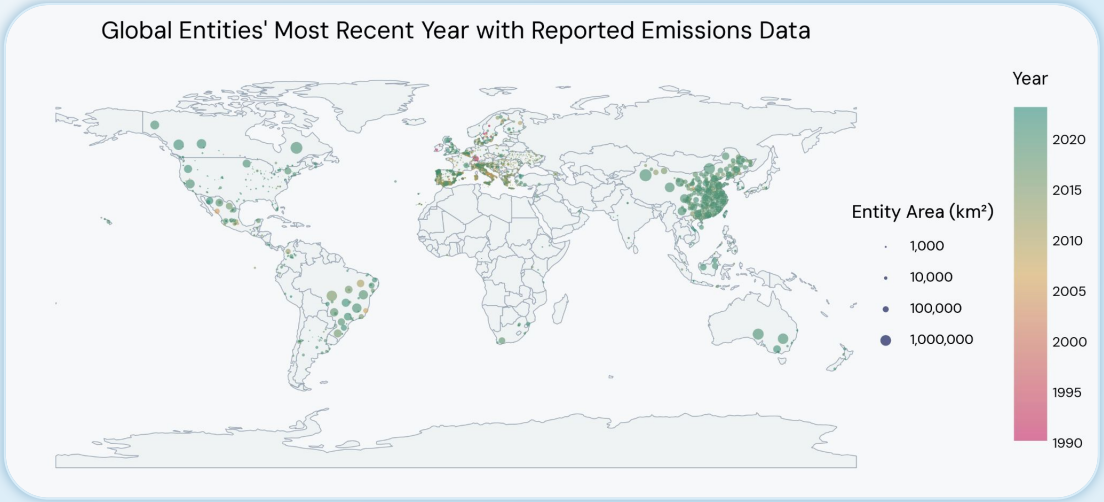
The emissions data are also asynchronous (the most recent year of emissions data is not the same for each entity)

us_cities_emissions.png

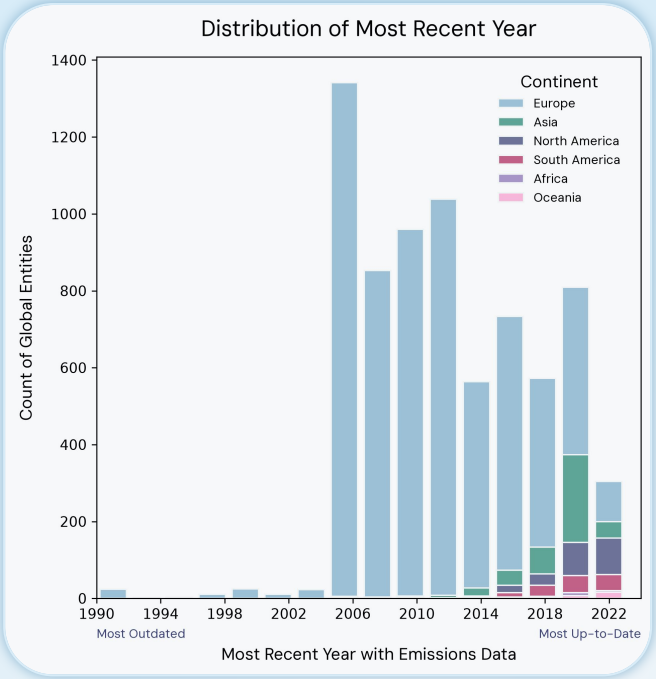


Subsetting to a single country (e.g., the U.S.) and `entity_type` *(e.g., cities) can eliminate redundancies and biases (other than* `inv_year` *still being asynchronous).*

Emissions Data Recency



global_emissions_recency_map.png



global_emissions_recency_hist.png

Years with emissions data (`inv_year`) fall on a wide range. Most entities have at least one year of data from the past five years, but I found 47 entities whose most recent data is from the 1990s.

This could introduce biases if users blindly use the emissions data.

Entity Type Bias



global_entity_types.png

Differences in `entity_type` reporting between regions potentially introduces bias to interregional or global comparisons.

This figure also illustrates the abundance of emissions data from European cities and an overall lack of data from the Global South.

Data Quality & Potential Updates

There are minimal issues with data quality in the dataset's most important features. I did find a few potentially erroneous values in the `inv_year` *and* `total_emissions` *columns (see my code), but those could be a misunderstanding on my part.*

Primary Limitations Encountered		Potential Solutions
Emissions data for different entities can be from different years, making one-to-one comparisons less valid.	→	If possible, source more data from recent years.
The most common `entity_type` differs between regions, meaning either: (1) if all entity types are included, global graphs will have overlapping data; or (2) if we filter for only one type of entity, some countries will lose a greater proportion of their entities (datapoints) than others.	→	If possible, source data from cities in countries with mostly regional emissions data, and data from regions/subregions in countries with mostly city-level emissions data.
Different entities are biased toward one of the two emissions units (tCO_2 vs. tCO_2e), meaning either: (1) entities that measure in tCO_2e are biased toward higher `emissions` values; (2) if we filter for only one measure, some countries will lose a greater proportion of their entities (datapoints) than others.	→	Pivot measures of tCO_2 and tCO_2e into two separate `total_emissions` columns so they aren't accidentally compared. Source both measures when possible in order to reduce missing values.

Data Maintenance

When updating the data files with new data:

- Ensure there aren't duplicate data for the same entity and year (`climactor_id` and `inv_year`)
- Check consistency of column names, column types, and value formatting
  - Coordinates are especially important and could easily be formatted differently
- Monitor important features for number of missing values

Thank you for your time and consideration.

GitHub Repository: github.com/machristians/research_specialist_test

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